NORMAN BUI

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Education

Carleton University | *Bachelor of Computer Science (Honours),* ***3.9/4.0 GPA*** Graduated Spring 2026

- Coursework: Data Structures & Algorithms, Operating Systems, OOP, Programming Paradigms, DBMS
- Awards: Dean's Honour List every year; President's, Henry Marshall Tory, and Harry S. Southam Scholarships

Experience

Canadian Tire Corporation | *Software Developer* July 2026 - Present

- Built an internal tool that reads a service's source code and writes its documentation, across **12** languages
- Writes **3** documents off the same facts: a developer guide, an architecture page, and a plain-language page for non-engineers
- Rewrote it from TypeScript to Rust with identical output, shipped as one program for **6** platforms
- Audited **1,694** claims (**1.9%** wrong or misleading) and cut the cost of each run **38x** by testing AI models

Electronic Arts (EA) | *Software Developer Intern* May 2025 - Aug 2025

- Raised automated performance test pass rates from **35% to 100%** by fixing broken logins and setup steps
- Integrated **10+** game features into load tests for high-traffic performance coverage
- Reviewed an AI system that plays the game to find bugs, recommending fixes for long test-run reliability

Autodesk | *Software Developer Intern* Jan 2025 - Apr 2025

- Built systems that find and fix missing manufacturing data in Autodesk Fusion, with a repair step **8x** faster
- Shipped services that check and repair customer data automatically, with **100%** test coverage
- Closed security gaps in how sensitive customer manufacturing data was accessed

Kongsberg Geospatial | *Software Developer Intern* May 2024 - Aug 2024

- Built live mapping demos for aviation, defense, and government clients as the only developer on the team
- Combined several terrain and mapping data formats into one live view with real-time object tracking
- Automated airspace classification, removing a manual sorting step

Health Canada | *Data Scientist Intern* Sep 2023 - Dec 2023

- Rebuilt the system that gathers chemical data, more than doubling usable sources with **100%** test coverage
- Built models that predict how toxic a chemical is likely to be, supporting risk assessments
- Made the department-wide chemical search more reliable, raising successful lookups by **20%**

NAV CANADA | *Software Developer Intern* May 2023 - Aug 2023

- Reworked an internal mapping tool to use **60%** less storage and analyze data **30%** faster
- Fixed display bugs in air traffic control software, making it **16%** smoother to use
- Automated network setup, cutting manual configuration work by more than **95%**

Projects

Orbital Refuelling Simulator | *Python, Streamlit, scikit-learn, pytest* github.com/nbui23/orbital-refuelling

- Simulated a **410-second** spacecraft refuelling mission across **9** phases, with **9** kinds of failure to catch
- Combined fixed safety limits with machine learning that flags subtler drift, keeping the two clearly separate
- Ranked which sensor readings drove each alert, so an operator can see why it fired

Skills

Languages: Python, TypeScript, JavaScript, C++, C#, Kotlin, Java, SQL, HTML/CSS, PHP
Frameworks & Runtimes: Next.js, Node.js, Express.js, Electron, Qt, React Native, Expo, Spring Boot, FastAPI, Flask, Streamlit, .NET, ASP.NET
AI / ML / Data: LLMs, RAG, LangGraph, LangChain, FAISS, Ollama, SBERT, scikit-learn, IsolationForest, pandas, NumPy, matplotlib, PyTorch, ONNX Runtime
Databases & APIs: PostgreSQL/PostGIS, pgvector, MySQL, SQLite, MongoDB, Firebase, Neo4j, Prisma, GraphQL
Tools & Practices: Git, AWS, CI/CD, Docker, Docker Compose, Ubuntu, TDD, pytest, Mocha, Chai, Sinon, unittest, BS4, Selenium, Cucumber, PowerShell, Postman, Agile, SDLC
Spoken Languages: English, French, Vietnamese